

A Causal-Guided Multimodal Large Language Model for Generalized Power System Time-Series Data Analytics

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Abstract—Power system time series analytics is critical in understanding the system operation conditions and predicting the future trends. Despite the wide adoption of Artificial Intelligence (AI) tools, many AI-based time series analytical models suffer from task-specificity (i.e., one model for one task) and structural rigidity (i.e., the input-output format is fixed), leading to limited model performance and resource waste. In this paper, we propose a Causal-Guided Multimodal Large Language Model (CM-LLM) that can solve heterogeneous power system time-series analysis tasks. First, we introduce a physics-statistics combined causal discovery mechanism to capture the causal relationship, which is represented by a graph, among power system variables. Second, we propose a multimodal data preprocessing framework that can encode and fuse text, graph and time series to enhance the model performance. Last, we formulate a generic “mask-and-reconstruct” paradigm and design a dynamic input-output padding mechanism to enable CM-LLM to adapt to heterogeneous time-series analysis tasks with varying sample lengths. Simulation results based on open-source LLM Qwen and real-world datasets demonstrate that, after simple fine-tuning, the proposed CM-LLM can achieve satisfying accuracy and efficiency on three heterogeneous time-series analytics tasks: missing data imputation, forecasting and super resolution.

Index Terms—Power system time-series analytics, large language model, causal relationship, multimodal data preprocessing, model generalization

NOMENCLATURE

Abbreviations

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers

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CM-LLM

DAG
DAUG
DGP
DTW
FID
GCN
GNN
LLM
LoRA
LSTM
MAE
MLP
NLP
PC
PV
RMSE
SAE
SFT
SOC
TSFM

Causal-Guided Multimodal Large Language Model

Directed Acyclic Graph
Directed Acyclic Unweighted Graph
Dense Graph Propagation
Dynamic Time Warping
Fréchet Inception Distance
Graph Convolutional Network
Graph Neural Network
Large Language Model
Low-Rank Adaptation
Long Short-Term Memory
Mean Absolute Error
Multi-Layer Perceptron
Natural Language Processing
Peter-Clark
Photovoltaic
Root Mean Square Error
Stacked Auto-Encoder
Supervised Fine-Tuning
State of Charge
Time-series Foundation Models

Parameters

B	Batch size used in training
D_{hidden}	Hidden dimension of the embedding vectors
E	Number of variables in the multivariate time series
$\mathcal{E}$	Set of edges in the causal graph
L	Length of the complete time series sequence
L_{fix}	Predefined fixed length for input standardization
L_{ts}	Length of the time series segment
L_{txt}	Length of the instruction prompt sequence
L_{valid}	Valid sequence length of a sample
r	Rank hyperparameter in LoRA adaptation
$\mathcal{V}$	Set of nodes in the causal graph
W_0	Pretrained weight matrix in the LLM
λ_1	Weighting coefficient
μ, σ^2	Mean and variance parameters for the normal distribution used in masking

Variables

A, B	Trainable low-rank matrices in LoRA
E_{causal}	Causal embeddings encoded from the graph
E_{fused}	Fused embeddings combining temporal and causal features
E_{input}	Final concatenated input embeddings
E_{ts}	Temporal embeddings of the time series data
E_{txt}	Prompt embeddings derived from textual instructions

f_{θ}	Mapping function of the model
G_e	Explicit causal graph structure
G_c	Complete causal relationship graph
H_{output}	LLM hidden state sequence
I_{txt}	Input instruction text sequence
L_{acc}	Accuracy loss
L_{mask}	Mask reconstruction loss
$\mathbf{M}$	Binary mask matrix
$\mathbf{X}$	Multivariate time series data matrix
$\tilde{\mathbf{X}}$	Masked input time series data
$\hat{\mathbf{Y}}$	Reconstructed time series output

I. INTRODUCTION

DRIVEN by global energy transition and the strategic pursuit of carbon neutrality, modern power systems are undergoing a profound transformation. The large-scale integration of renewable energy sources and flexible loads has not only increased the complexity of grid dynamics, but also generated massive volumes of high-dimensional time-series data enabled by advanced metering infrastructure [1]. As such, extracting operational insights from the large-scale, noisy, and incomplete datasets has become a key challenge in modern power systems. These tasks can be collectively referred to as time-series data analysis in power systems. For example, data imputation aims to ensure data integrity by recovering missing or corrupted values; short-term forecasting provides decision support for day-ahead scheduling by predicting future system states; super-resolution enhances temporal resolution to support fine-grained modeling and control.

To handle the time series data analysis problem, Artificial Intelligence (AI), particularly deep learning models, has gained increasing attention in power systems, demonstrating significant achievements in enhancing system reliability and efficiency [2]–[5]. However, as illustrated in Fig. 1, most existing methods suffer from the following two limitations, which impair their real-world implementation:

- **Task specificity:** Most AI-based models are designed for a specific task with tailored architectures. However, some diverse tasks (such as data imputation, super-resolution, and forecasting) are intrinsically interconnected as they stem from identical physical mechanisms. Isolating these tasks into separate models ignores the shared physical patterns, preventing the model from learning a better and comprehensive representation of system dynamics and leading to resource waste.
- **Structural rigidity:** While some architectures (e.g., recurrent neural network) can theoretically process time series of varying lengths, standard deep learning pipelines typically rely on fixed input-output dimensions. Such structure creates difficulties when handling flexible queries with distinct input-output formats (e.g., switching from a missing data restoration task to a forecasting task). Consequently, existing methods usually require architectural modification or retraining to adapt to different tasks.

In recent years, the emergence of foundation models, especially large language models (LLMs), offers a promising new paradigm for solving the above two limitations. Based on a single pre-trained backbone, these models demonstrate unprecedented generalization capabilities that can support a wide range of downstream tasks, paving the way for realizing a truly generalized time

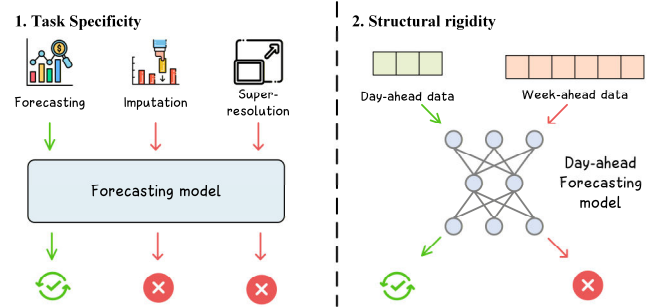


Fig. 1. Illustration about task specificity and structural rigidity problems for traditional AI models.

series analysis framework in power systems [6]–[8]. Currently, there are three technical mainstreams of leveraging foundation models for domain-specific general-purpose modeling:

- **Domain-specific pretraining:** This approach involves pre-training specialized backbone models from scratch using large-scale domain data. For instance, Tu et al. develop PowerPM, a foundation model pre-trained on diverse power system time-series data [7]. Wang et al. proposed a unified numerical model for smart meter analytics [9]. While such models are highly customized and capable of generating professional results, training them requires massive amounts of domain-specific data and extensive computational resources, often limiting their parameter scale compared to leading commercial LLMs.
- **Adaptation of Time Series Foundation Models (TSFMs):** Advanced time-series foundation models like TimeGPT [10] and Chronos [11] are built upon massive-scale numerical pre-training to achieve remarkable data analytics capabilities. Although these TSFMs capture complex time-series patterns and support task-specific fine-tuning, they lack the multimodal ability to incorporate non-numerical information (e.g., contextual operational records or social events, physical system models, etc.) that can influence power system dynamics, limiting their performances.
- **Adaptation of Pretrained LLMs:** This approach is based upon the state-of-the-art general-purpose LLMs that have been pretrained at great cost, such as Qwen, Llama, and ChatGPT. These models have powerful capabilities in sequence modeling, instruction following and multimodal processing, which can be adapted to time-series analysis tasks through techniques such as supervised fine-tuning. Recent studies [12], [13] demonstrate that the self-attention mechanisms in pretrained LLMs possess architectural universality, enabling them to extract temporal features from numerical sequences. Unlike time-series foundation models such as TimeGPT, LLMs leverage multimodal integration capabilities to fuse unstructured information with numerical inputs to capture complex dependencies [14], [15]. This approach not only significantly lowers the development costs of domain-specific LLMs [16] but has also been proved effective in time-series tasks like wind power forecasting [15].

Note that compared with economic and social time series such as stock price, GDP growth, etc., one distinct feature of multivariable power system time series is that they are governed by strict physical laws instead of noisy correlations. For instance, the power flow among the grids follows Kirchhoff's law, and the State-of-Charge (SOC) of a

battery can be calculated by integrating current over time with respect to time. In such cases, existing statistical-based foundation models, if not regulated by physical laws, may lead to errors when processing multivariate power system time series. Therefore, it is necessary to incorporate the physical laws into existing LLMs to formulate a physics-statistics combined model when handling power system time series.

In this paper, we develop a Causal-guided Multimodal LLM (CM-LLM) for physics-aware and generalized power system time series data analysis purposes. The physical laws that govern the time-series variables in power systems are described as *causal relationships* and are correspondingly modeled by causal graphs. Then the causal graphs, together with numerical time-series data and textual prompts reflecting task information, are combined and fed into multimodal LLMs to generate results with physical consistency.

Our main contributions are summarized as follows:

1. A physics-statistics combined causal discovery mechanism: We propose a novel causal discovery mechanism that can effectively capture the causal relationship among power system time series variables. This mechanism leverages the physical connections among variables as the backbone of the causal relationship, which is then optimized by the Peter-Clark (PC) algorithm based on field measurements. As such, this mechanism has good interpretability and reliability in causal structure learning, providing an important modality input to the proposed CM-LLM model.

2. A unified multimodal data preprocessing architecture: We design a data preprocessing architecture that can unify multimodal data inputs, including textual instructions, graph structures (representing causal relationships) and multivariate time series. In this architecture, each modality is encoded into vectors respectively, followed by specially-designed data fusion mechanism. As such, this data preprocessing architecture can enable the proposed CM-LLM model to gain a deep understanding about complex power system patterns and tasks.

3. A dynamic input-output padding mechanism: We propose a dynamic input-output padding mechanism, including dynamic batch-wise padding and sequence-aware loss computation, to handle the varying input-output sequence lengths of different time series analysis tasks. As such, the structural rigidity issue can be solved without sacrificing model accuracy.

The remainder of this paper is organized as follows: Section II reviews the related works; Section III presents the methodology; Section IV demonstrates the case study results; and Section V provides the conclusions.

II. RELATED WORK

A. Power System Time-Series Data Analysis

Time-series data analysis enhances operators' understanding of grid situation and serves as the foundation for decision-making process. Currently, most time-series analysis models are designed for specific tasks and are trained in end-to-end fashion on small-scale datasets [3]. For example, to capture temporal dynamics, recurrent neural networks, such as Long Short-Term Memory (LSTM)[17] and gated recurrent units [4], have been widely adopted in forecasting tasks. Generative models [18], particularly those based on generative adversarial networks and diffusion

processes, have also been implemented in time-series analysis tasks such as forecasting [19], data imputation [20] and super-resolution [21] due to strong distribution-learning ability. In addition, temporal convolutional networks [5], Transformer-based architectures [22] and Kolmogorov-Arnold Networks (KAN) [23] have demonstrated superiority in modeling power system data dependencies. Considering that power system is essentially a multivariate system, researchers have introduced auxiliary information such as graph structure [24] and cross-variable correlations [25] to enhance the time-series analysis performances. While these approaches have achieved promising results on individual tasks, they often lack generalization and adaptation capabilities to heterogeneous tasks (i.e., task specificity issue) and are constrained by fixed input-output configurations (i.e., structural rigidity issue). To address these limitations, the proposed method utilizes a unified "mask-and-reconstruct" paradigm and a dynamic padding mechanism to handle different time series analysis tasks. Unlike traditional models requiring distinct architectures for specific tasks, this framework allows a single backbone to process diverse time-series tasks, thereby mitigating the constraints of task specificity and structural rigidity.

B. LLM Applications in Power Systems

LLMs have recently gained increasing attention in power systems due to their versatile modeling and generalization capabilities [6]. Based on the model pretraining strategy, Tu et al. establish a hierarchical-temporal model pretrained on electricity time series data, achieving superior performance across downstream tasks [7]. Zhu et al. introduce a physics-informed graph transformer that leverages grid topologies and dynamic mixture-of-expert layers [26]. Hu et al. propose BERT-PIN, a framework that leverages natural language processing techniques for power load restoration by encoding time-series data as linguistic tokens in a learned "language" of consumption patterns [27]. Other researchers adapt existing pre-trained LLMs through techniques such as fine-tuning [28], in-context learning [29], and retrieval-augmented generation [30]. In addition, some researchers have explored the use of LLMs as intelligent assistants to support decision-making processes. For instance, LLMs have been employed to recommend predictive models [31], generate code [32], [33], produce reinforcement learning strategies [34], or provide suggestions for optimal operational adjustments [8]. However, while some frameworks incorporate textual prompts with numerical data, they lack explicit causal integration of structural physical systems. In contrast, the proposed method establishes a multimodal architecture that deeply fuses textual instructions, causal graphs, and time-series data, aligning generative results with physical mechanisms and task instructions.

C. Causal Analysis in Power Systems

Causal relationships represent intrinsic and invariant dependencies among variables that remain stable across different conditions [35]. Because power system is a typical multivariate physical system, uncovering such causal structures can provide a more reliable and interpretable basis for data analysis [36], [37], [38]. For example, Zhao et al. propose CEDAN, a causality-enabled domain adaptation network that extracts invariant causal features from pre-trained source PV data, enabling unsupervised zero-label forecasting for new photovoltaic sites [39]. Jiang et al. show

that causal meta-learners, pre-trained on debiased observational data via propensity trimming, can accurately estimate HVAC impacts on building thermal dynamics [40]. While these studies demonstrate the utility of causal analysis, they typically rely on data-driven discovery algorithms or fixed physical topologies. In contrast, the proposed method implements a physics-statistics combined causal discovery mechanism. By utilizing physical laws to define the initial skeleton and statistical tests to refine connections, this approach constructs a causal graph that maintains physical consistency while adapting to observational data.

D. Research Gaps

Despite advancements in power system time-series analysis, three critical research gaps are identified. First, establishing a unified paradigm capable of handling heterogeneous tasks without task-specific structural modifications constitutes an unresolved challenge. Second, multimodal data fusion strategies based on LLMs, particularly the effective integration of text, graph structures, and time-series data, require further exploration. Third, the explicit embedding of inherent physical relationships into the generation process to enhance the physical consistency remains an under-explored area.

III. METHODOLOGY

In this section, we select three typical time-series analysis tasks: data imputation, forecasting and super-resolution, as examples to demonstrate how the proposed method can unify them into a common representation. Next, we introduce the proposed CM-LLM framework, which enables generalized time series analysis for power systems through a single training process.

A. Problem Formulation

The adopted "mask-and-reconstruct" paradigm provides a generalized framework for power system data analysis. Theoretically, any task involving the estimation of unmeasured or corrupted variables from available data can be formulated within this approach. However, to systematically evaluate the multimodal causal mechanism, the scope of this paper is explicitly defined as multivariate time series reconstruction. Accordingly, data imputation, forecasting, and super resolution serve as the representative tasks to verify the proposed framework.

As illustrated in Table I and Fig. 2, despite the differences among the tasks of data imputation, forecasting and super-resolution, their core objective is fundamentally the same: to infer and reconstruct unknown information by learning the patterns from the observed time series.

If we treat the unknown regions as masked segments, the three different tasks can be uniformly formulated as a "mask-and-reconstruct" process. In the case of imputation, the mask corresponds to missing segments in the time series, and the goal is to reconstruct these missing parts using contextual information. For forecasting, the mask covers all future time steps, and the model infers future trends based on historical observations. In super-resolution, the mask represents the implicit high-resolution points to be recovered between low-resolution measurements.

Given that power systems are typically multivariate in nature, we can thus obtain a unified problem formulation for these three tasks as follows:

Denote a complete multi-variable time series data matrix

$\mathbf{X}$, including system and explanatory variables as

$$\mathbf{X} = \begin{bmatrix} x_1^1 & x_1^2 & \cdots & x_1^L \\ x_2^1 & x_2^2 & \cdots & x_2^L \\ \vdots & \vdots & \ddots & \vdots \\ x_L^1 & x_L^2 & \cdots & x_L^L \end{bmatrix} \quad (1)$$

where E is the number of variables and L is the length of the time series. We define a mask matrix $\mathbf{M}$, which consists of binary values, 0 and 1. The size of $\mathbf{M}$ is similar to $\mathbf{X}$.

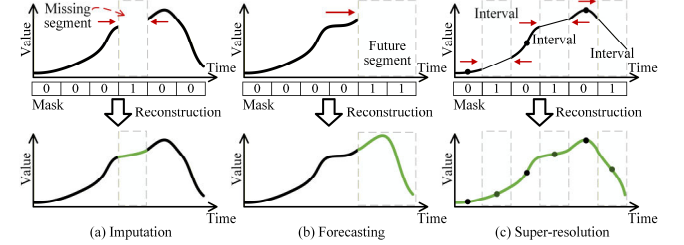


Fig. 2. The "mask-and-reconstruct" generic representation paradigm applied to different tasks. (a) Imputation: recovering missing segments in damaged data. (b) Forecasting: predicting future values based on historical context. (c) Super-resolution: reconstructing fine-grained time series from low-resolution coarse samples (sparse points), thereby increasing the granularity of the data.

TABLE I
SUMMARY OF THREE TASKS

Task	Imputation	Forecasting	Super-resolution
Objective	Restore data integrity	Infer future states	Increase data granularity
Unknown position	Missing segments within a known time range	Future segments beyond the known time range	Samples between low-resolution points
Data context	Bi-directional	Uni-directional	Bi-directional

In the mask matrix, we designate segments to be marked as 1. Therefore, input data $\tilde{\mathbf{X}}$ can be viewed as a masked version of the underlying complete multivariate time series, where certain entries are missing or unobserved:

$$\tilde{\mathbf{X}} = (\mathbf{I} - \mathbf{M}) \circ \mathbf{X} \quad (2)$$

where $\circ$ denotes the Hadamard product (element-wise multiplication). Thus, the problem can be described as:

$$\hat{\mathbf{X}} = f_{\theta}(\tilde{\mathbf{X}}), \text{ with } \text{Loss}(\hat{\mathbf{X}}, \mathbf{X}) \quad (3)$$

where f_{θ} is the mapping function. The goal is to find a function, f_{θ} , to reconstruct the complete multi-variable time series data.

B. CM-LLM Model Architecture

As illustrated in Fig. 3, the proposed CM-LLM model comprises three fundamental stages: multimodal data preprocessing, multimodal information fusion, and CM-LLM inference. Through the Supervised Fine-Tuning (SFT), the model can understand the time series patterns and is able to transform the multi-source heterogeneous inputs to diverse outputs. In the multimodal data preprocessing stage, we collect input data from multiple modalities and embed them into an LLM-oriented pattern. This is because LLM is originally designed for Natural Language Processing (NLP) tasks, which expects inputs in the form of tokenized word embeddings. Therefore, during the input adaptation process, we need to generate embeddings for each modality accordingly. First, textual data are derived from natural language descriptions of tasks and scenario instructions, which are then converted into prompt embeddings. Second, raw power system time-series data are processed through a masking mechanism and transformed into unified temporal

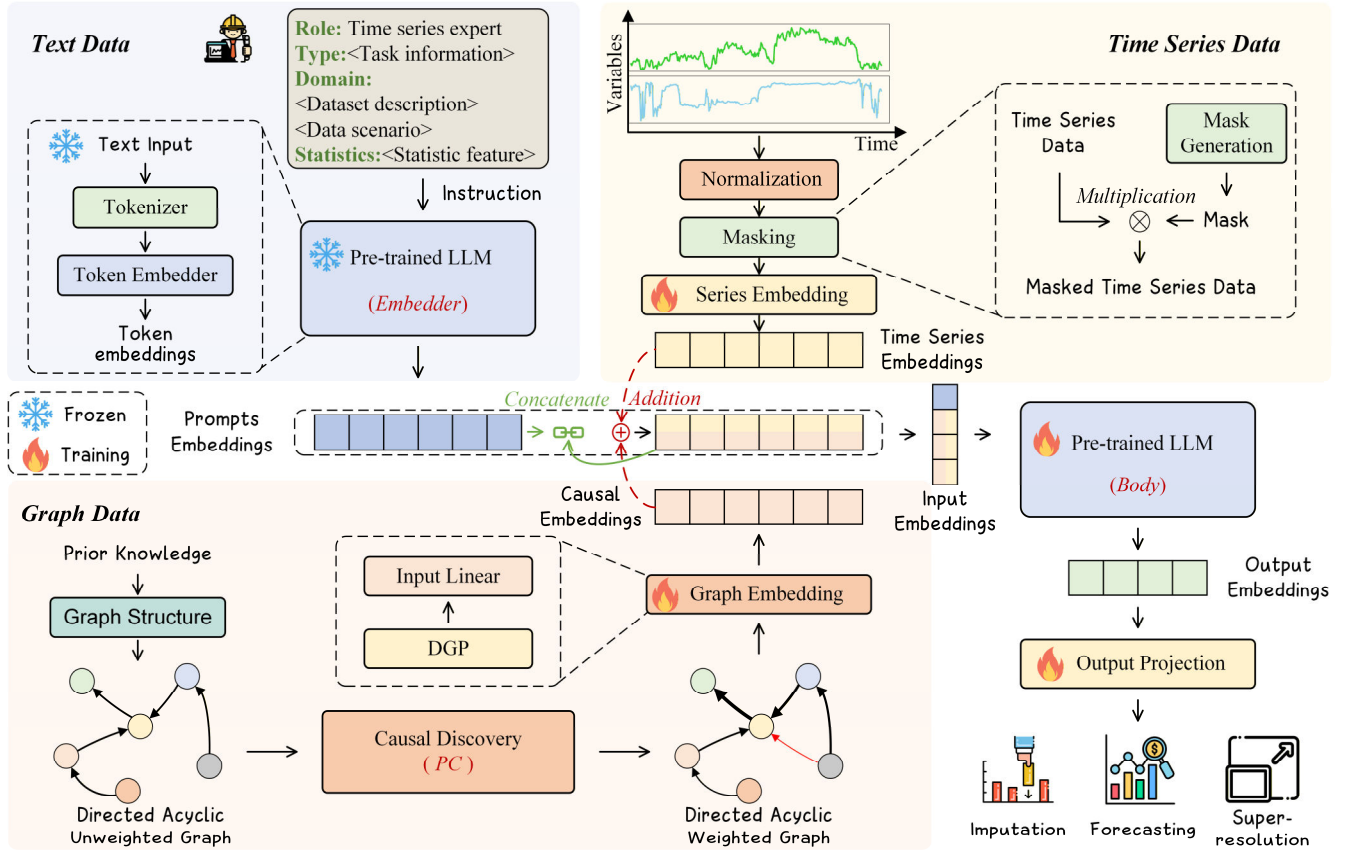


Fig. 3. The model framework of CM-LLM. The model integrates three modalities: text, time-series, and graph data. (Top-left) Text data: Prompts are constructed using a standardized template enriched with statistical features and domain knowledge to provide semantic context. (Bottom-left) Graph data: A causal graph is discovered via the PC algorithm to capture structural dependencies. (Center) Multimodal fusion: The fusion process consists of two stages: first, causal embeddings are injected into time-series embeddings via element-wise addition; second, text embeddings are concatenated as a prefix to the fused sequence. Finally, the unified representations are fed into the trainable LLM backbone for downstream tasks (imputation, forecasting, and super-resolution).

embeddings. Third, prior causal graphs are refined using the PC algorithm and subsequently encoded into causal embeddings via a Graph Neural Network (GNN).

The second stage of the framework is multi-modal information fusion, wherein embeddings derived from diverse data modalities are systematically aligned and effectively integrated. First, temporal embeddings and causal embeddings are combined through an element-wise summation operation, aiming to enrich the temporal feature representation with encoded variable interactions. The resulting embedding vector is then concatenated with the prompt embeddings, which are generated from task-specific textual instructions, to construct a comprehensive sequence representation. Such integrated representation not only encapsulates rich contextual and semantic information but also incorporates structured guidance for the target task, thereby facilitating the model's reasoning and decision-making capabilities within the given application scenario.

In the final CM-LLM inference stage, the fused sequence is input into a pre-trained LLM, which functions as the core reasoning module for capturing complex nonlinear dependencies and long-term temporal dynamics. The hidden state sequence generated by the LLM is then extracted. Using the known length of the instruction prompt, a slicing operation isolates the segment corresponding to the time series. Subsequently, this segment is mapped to the target output dimension through a linear projection layer.

In the following sections, we will introduce the above three main components in detail.

C. Multimodal Data Preprocessing

1) Textual Modality: Instruction Prompts

Prompt engineering optimizes LLM performance by constructing specific instructions. As shown in Fig. 3, the proposed architecture comprises two components: basic contextual description and statistical input analysis. The former establishes a semantic framework through task type, role assignment, dataset description, and data scenario definition. The latter extracts time-variant characteristics from input sequences and encodes numerical features into linguistic description. The integration of these two components enables the model to execute complex temporal analysis with semantic coherence and context awareness.

2) Temporal Modality: Multivariate Time Series

Multivariate time series data constitutes the main part of the SFT dataset. The processing procedure is as follows:

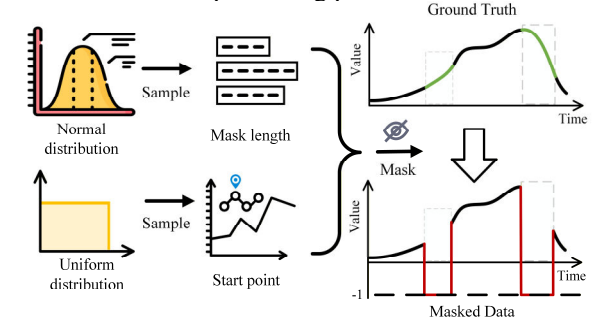


Fig. 4. The masking method for data imputation.

First, each temporal variable is individually normalized using normalization, transforming the data to lie within the interval $[0,1]$ to obtain the standardized dataset $\mathbf{X}$.

Second, considering that mask positions are different in tasks, 3 strategies for generating mask matrix $\mathbf{M}$ are designed for different tasks. Among the tasks considered, imputation involves the most complex masking strategy. Specifically, as shown in Fig. 4, we randomly select continuous time segments in $\mathbf{M}$ and mark them as missing (i.e., assigned a mask value of 1), where the length of each masked segment is sampled from a normal distribution. The parameters of the distribution are tunable and can be adapted according to the specific characteristics of the dataset in use. The process can be formulated:

$$\ell = \max\left(0, \left\lfloor \mathcal{N}(\mu, \sigma^2) \right\rfloor\right); s \sim \mathcal{U}(1, L - \ell_i + 1) \quad (4)$$

$$M[s:s + \ell_i - 1] \leftarrow 1 \quad (5)$$

where ℓ is the length of the missing data segment in imputation task, $\mathcal{N}(\mu, \sigma^2)$ is the normal distribution with a mean of μ and a variance of σ^2 , s is the starting point of missing data segment, $\mathcal{U}$ is a uniform distribution and L is the length of total segment. Unlike imputation where missing segments can occur anywhere in the time series, the endpoint in a forecasting task is fixed, and the forecasting window is also fixed. In this case, we only need to determine the length of the future window to be predicted. The values within the prediction window are masked out by the mask matrix $\mathbf{M}$. For super-resolution, the generation of the mask matrix $\mathbf{M}$ is straightforward: all positions in the mask matrix corresponding to the intermediate points between low-resolution observations are set to 1. The effects of mask matrix are two-fold: (1) It is used to mask true values in the input data, thereby facilitating the construction of SFT training pairs; (2) It serves to identify the masked positions and facilitates the calculation of the loss function by focusing on the relevant regions of the input.

Third, based on the mask matrix, all elements in the dataset $\mathbf{X}$ that correspond to positions marked as 1 in the mask are replaced with -1. It is worth noting that, since the data in $\mathbf{X}$ have already been normalized to the range $[0, 1]$, the value -1 serves as a distinct and easily identifiable indicator of masked positions.

$$\tilde{\mathbf{X}} = (\mathbf{I} - \mathbf{M}) \circ \mathbf{X} + (-1) \cdot \mathbf{M} \quad (6)$$

3) Graph Modality: Causal Relationship

Prior knowledge refers to established domain expertise and governing physical laws. This knowledge is obtained directly from fundamental power system mechanisms rather than statistical data estimations, such as power balance equation. The term physics denotes these invariant structural dependencies and conservation laws. The established physical laws form the explicit prior causal graph. This graph acts as a strict structural constraint to prevent erroneous edge generation caused by data noise. We encode this knowledge as a Directed Acyclic Unweighted Graph (DAUG), where nodes represent system variables and directed edges denote these causal dependencies.

Rooted in stable physical constraints, the DAUG serves as a foundational causal hypothesis, providing a structured, computationally tractable framework derived from domain expertise. To refine this initial structure and quantify causal strengths, we employ a physics-statistics mechanism. This approach utilizes the PC algorithm, which applies conditional independence testing to observational data to infer a more comprehensive causal graph.

The core procedure of the mechanism proceeds as shown in Fig. 5 and Algorithm 1:

First, a fully connected undirected graph U is initialized, where every pair of variables is connected by an undirected edge. Then, for each pair of variables, the Fisher Z conditional independence tests are conducted under varying conditioning sets. If a pair of variables is found to be conditionally independent given a certain set, the corresponding edge is removed. Notably, explicitly defined causal edges G_e derived from prior knowledge are preserved without undergoing such tests. For the remaining edges, the algorithm further evaluates the predictive power between variable pairs using linear regression models. Specifically, variable u is used to predict v , yielding the coefficient of determination R_{uv}^2 , and vice versa for R_{vu}^2 . By comparing these two values, the direction of the edge is assigned from the variable with higher predictive power to the one with lower predictive power.

Throughout this process, all edges representing explicitly defined causal relationships based on prior knowledge are retained in the final graph without participating in the conditional independence tests. This ensures effective utilization of prior knowledge while also improving computational efficiency. The resulting output is a Directed Acyclic Graph (DAG) that incorporates both prior-guided and data-validated causal relationships. To further quantify the significance of each causal edge, we compute weights for every directed edge ($u \rightarrow v$) in the resulting DAG. Specifically, for the edge ($u \rightarrow v$), we construct a conditioning set $\mathbf{Z}$ consisting of other parent nodes of v , and calculate the absolute value of the partial correlation coefficient. The absolute value of this partial correlation serves as the weight of the edge, reflecting the strength of the causal influence of u on v .

To handle dynamic causal relationships, our framework employs a sample-level graph construction strategy. While the physical topology G_e rarely changes, any structural shifts are identified during the data preprocessing stage. These changes are immediately incorporated into the prior knowledge, allowing the causal graph G_c to be dynamically instantiated for each specific sample. Furthermore, the self-attention mechanism provides a second layer of adaptation by adjusting the influence of causal edges based on real-time data patterns. This dual-layer approach ensures the model remains aligned with both abrupt structural changes and gradual shifts in system dynamics.

To ensure reliability against data noise, specific safeguards are integrated into the construction mechanism. First, the explicit physical causal graph G_e serves as a stable anchor. Verified physical connections are preserved without statistical testing to avoid the loss of true dependencies. Second, a strict significance threshold is applied to conditional independence tests during structure learning. This conservative approach retains only edges supported by strong evidence and filters out weak correlations from measurement noise. Third, the causal structure is formulated as a continuous weighted matrix rather than a binary constraint. Edges associated with noise typically carry lower weights and their influence is naturally attenuated. Finally, the attention mechanism of the backbone network acts as a dynamic filter. It adjusts the importance of structural priors during training to suppress artifacts conflicting with observed data patterns.

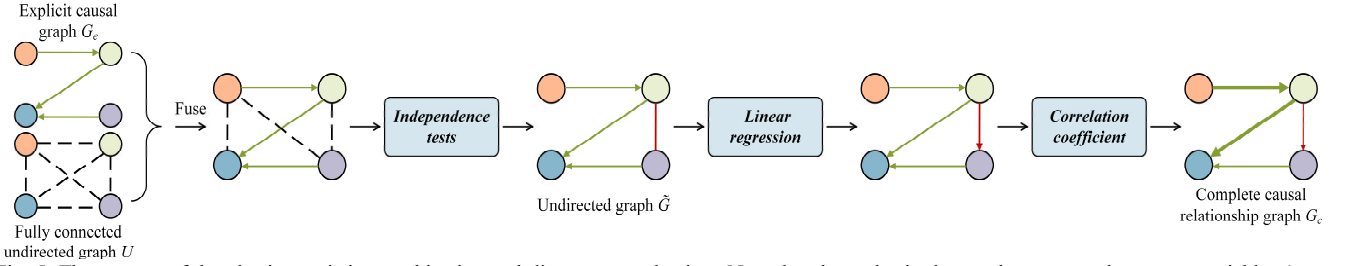


Fig. 5. The process of the physics-statistics combined causal discovery mechanism. Note that the nodes in the graph represent the system variables (e.g., active power, voltage, or meteorological factors) rather than individual sampling points. Green and red edges denote physical and statistical causal relationships, respectively, while the edge thickness corresponds to the causal strength.

Algorithm 1: Physics-statistics combined causal discovery

Input: Explicit causal graph G_e , supervised fine-tuning dataset D_s .
Output: Complete causal relationship graph G_c .

Procedure:

- 1 Initialize fully connected undirected graph U .
- 2 **for** each edge $(u, v) \in U.edges()$ **do**:
- 3 **if** $(u, v) \in G_e.edges()$ **:**
- 4 **break**
- 5 **else**
- 6 Perform **Fisher Z** conditional independence test
- 7 **if** independent:
- 8 Remove the edge (u, v)
- 9 **else**
- 10 Remain the edge (u, v)
- 11 **end**
- 12 **end**
- 13 **end**
- 14 **return** undirected graph $\tilde{G}$
- 15 **for** each edge $(u, v) \in \tilde{G}.edges()$ **do**:
- 16 Fit $u \rightarrow v$ and $v \rightarrow u$ via linear regression.
- 17 Set direction by higher R^2 .
- 18 **end**
- 19 **return** directed graph $\hat{G}$
- 20 **for** each edge $(u \rightarrow v) \in \hat{G}.edges()$ **do**:
- 21 Compute **Pearsonr** $(u, v \mid \mathbf{Z})$ with parents $\mathbf{Z}$ of v .
- 22 Assign as edge weight.
- 23 **end**
- 24 **return** G_c

It is worth noting that the computational complexity of the linear regression step is strictly bounded. Because the system feature dimension is limited, and explicit physical priors along with Fisher Z tests heavily prune the search space, the regression is only performed on a highly minimized subset of unoriented edges.

4) *Fused Representation: Multimodal Embedding*

The three heterogeneous data modalities are encoded via different embedding approaches, and the corresponding representations are integrated into a cohesive input for the LLM. The whole instruction text I_{txt} is embedded by the tokenizer and token embedder of the LLM into prompt embeddings. Through fully connected layers, the time series data $\tilde{\mathbf{X}}$ is embedded into temporal embeddings. The embedding process can be formulated:

$$E_{txt} = f_{embed}(I_{txt}) \in \mathbb{R}^{B \times L_{txt} \times D_{hidden}} \quad (7)$$

$$E_{ts} = f_{ts}(\tilde{\mathbf{X}}) \in \mathbb{R}^{B \times L_{ts} \times D_{hidden}} \quad (8)$$

where f_{embed} is the token embedder, f_{ts} is the time series embedding layer, L_{txt} is the length of prompt, L_{ts} is the length of time series, and D_{hidden} is the hidden dimension of embeddings.

Spectral-based GNNs such as Graph Convolutional Networks (GCN) cannot directly process directed acyclic graphs [41]. These models are inherently designed for undirected graphs. The only feasible approach to apply them to directed graphs is to symmetrize the adjacency matrix thereby converting the directed graph into an undirected approximation [42]. To encode the causal relationship graph represented as a directed acyclic graph obtained from the prior oriented PC algorithm into informative vector representations, we employ a Dense Graph Propagation (DGP) module [43].

In DGP, the adjacency matrix is decomposed into ancestor connections and descendant connections according to the direction of the edges. The propagation is then computed separately for each direction to capture the distinct influence flows. An asymmetric normalization scheme is further applied to preserve the original directionality of the edges, ensuring that the causal structure of the graph is accurately represented during the message passing process.

The process of causal graph embedding can be formally formulated as follows. Notably, to align with temporal dynamics, the time series are incorporated as node features and jointly fed into the graph embedding layer.

$$G_c = (\mathcal{V}, \mathcal{E}), \text{ where } |\mathcal{V}| = F \quad (9)$$

$$E_{causal} = f_{gnn}(\tilde{\mathbf{X}}, G_c) \in \mathbb{R}^{B \times L_{ts} \times D_{causal}} \quad (10)$$

$$E'_{causal} = f_{graph}(E_{causal}) \in \mathbb{R}^{B \times L_{ts} \times D_{hidden}} \quad (11)$$

where G_c is the causal graph, $\mathcal{V}$ is the node set, $\mathcal{E}$ is the edge set, F is the number of variables, f_{gnn} is the DGP layer, and f_{graph} is linear layer.

D. *Multimodal Information Fusion*

To effectively integrate the modalities of time series data, task prompts, and causal graphs, we fuse the three distinct embeddings obtained through the data adaptation process into a unified feature vector that serves as the comprehensive input to the LLM. Specifically, we combine the time series embedding and the causal embedding via element-wise addition. This design is motivated by the observation that causal embeddings can be naturally interpreted as high-level features derived from the temporal dynamics of the data. The resulting fused embedding thus captures not only the intrinsic temporal evolution patterns within the time series but also the causal relationships among the variables.

In contrast to the causal embedding, which encodes intrinsic data characteristics, the prompt embedding represents textual descriptions regarding the task type and specific instructions. An effective prompt embedding enriches the input context, enabling the LLM to better comprehend the intended task. To this end, we adopt a

"Prompt-as-Prefix" strategy, in which the prompt embedding is concatenated with the time series embedding in a prefix fashion [3]. Given that our tokenizer is kept frozen during training, this approach ensures seamless compatibility and interpretability of the prompt embeddings by the LLM. The fusion process can be summarized as:

$$E_{fused} = E_{ts} + E'_{causal} \in \mathbb{R}^{B \times L_{ts} \times D_{hidden}} \quad (12)$$

$$E_{input} = \text{Concat}(E_{txt}, E_{fused}) \in \mathbb{R}^{B \times (L_{txt} + L_{ts}) \times D_{hidden}} \quad (13)$$

E. CM-LLM Inference

The multimodal fused embeddings E_{input} are fed as inputs to the LLM, which generates initial hidden state sequence H_{output} in an end-to-end process. The portion of the hidden state sequence corresponding to the time series is isolated by slicing, leveraging the known length of the instructional prompt L_{txt} . As shown in Fig. 3, the segmented hidden states H_{ts} are passed through a projection layer, f_{proj} , which projects them back to the original feature space to produce the final prediction $\hat{Y}$.

$$H_{output} = \text{LLM}(E_{input}) \in \mathbb{R}^{B \times (L_{txt} + L_{ts}) \times D_{hidden}} \quad (14)$$

$$H_{ts} = H_{output}[:, L_{txt} :, :] \in \mathbb{R}^{B \times L_{ts} \times D_{hidden}} \quad (15)$$

$$\hat{Y} = f_{proj}(H_{ts}) \in \mathbb{R}^{B \times L_{ts} \times F} \quad (16)$$

F. CM-LLM Fine-Tuning Process

1) Model Supervised Fine-Tuning

In SFT, LLM is trained with specially designed loss functions. The loss function includes 2 terms: the accuracy loss (L_{acc}) and the mask loss (L_{mask}), as shown in (17)–(18). λ_1 is the balanced weight. L_{acc} is a comprehensive loss to understand the changing pattern of time series data. L_{mask} employs mean squared error to minimize the point-to-point discrepancies in the mask segments.

$$\min \{L_{acc} + \lambda_1 L_{mask}\} \quad (17)$$

$$L_{acc} = \|\hat{Y} - \mathbf{X}\|_2^2, L_{mask} = \|\hat{Y} \circ \mathbf{M} - \mathbf{X} \circ \mathbf{M}\|_2^2 \quad (18)$$

To reduce the required training resources, we adopt a parameter-efficient fine-tuning approach known as Low Rank Adaptation (LoRA), which introduces a limited set of trainable parameters to adapt the pretrained model [44]. For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, its update is constrained through a low rank decomposition:

$$W = W_0 + T(W_0) = W_0 + BA \quad (19)$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$, with the rank $r \ll \min(d, k)$. This low rank approximation effectively reduces the number of trainable parameters while preserving the expressive capacity of the model during adaptation.

2) The Dynamic Input-output Padding Mechanism

It is worth noting that in different task settings, the lengths of input and output sequences can vary significantly. To effectively address the inherent variability in sequence lengths across multiple tasks in time series learning, our framework adopts a hierarchical strategy for both sequence processing and loss computation, as shown in Fig. 6. During the data preprocessing stage, all input sequences are standardized to a predefined fixed length L_{fix} through padding or truncation. This ensures structural consistency in the input space for subsequent deep learning components.

In contrast, the output sequence lengths may differ across tasks, while the output dimension of the projection layer remains fixed. To reconcile this discrepancy, we employ a

padding-and-slicing mechanism. Specifically, during label loading, a custom collate function dynamically pads the output labels within each batch to match the maximum sequence length L_{max}^{batch} present in that batch. This enables efficient batched computation. At the same time, the original valid sequence length L_{valid} for each sample is preserved as metadata and passed along with the batched data into the training pipeline. During the loss computation phase, we implement a sequence-aware approach: although the model output and padded labels may not align in length, the loss function uses L_{valid} to slice both predictions and ground truths, computing the error only over the valid portion of each sequence.

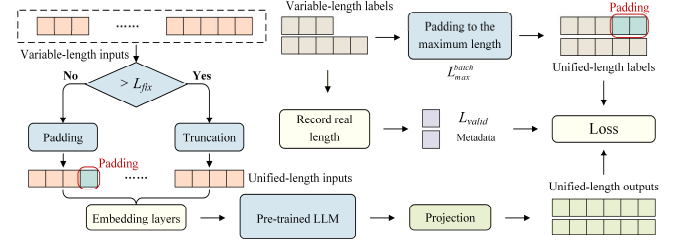


Fig. 6. The workflow of the dynamic padding process. To accommodate variable-length tasks, input sequences are standardized to a fixed length L_{fix} via padding or truncation. In parallel, labels are dynamically padded to the maximum length within a batch L_{max}^{batch} for efficient processing. Crucially, the sequence-aware loss mechanism utilizes the recorded valid lengths L_{valid} to slice the model outputs, ensuring that the loss is computed exclusively on valid data segments while ignoring padding values.

This strategy precisely isolates the influence of padding values on gradient backpropagation. As a result, our method enables efficient batch processing of variable-length sequences within a predefined maximum context window, overcoming the limitations imposed by the rigid structure.

IV. CASE STUDY

A. Experiment Settings

In this section, we conduct a comprehensive evaluation of the proposed LLMs across various tasks. The Qwen model is selected as the backbone pre-trained model due to its well-documented stability and broad applicability. To investigate the impact of model scale, we evaluate variants with approximately 0.6B, 1.7B, 4B, and 8B parameters. A series of ablation experiments are further carried out to analyze the contribution of different components and to better understand the model behavior.

Two distinct real-world datasets, referred to as dataset A and dataset B, facilitate the verification of the model. Dataset A consists of operational data collected from a photovoltaic (PV) power plant equipped with an energy storage system. The data have a temporal resolution of one minute. A total of 365 daily records constitute the supervised fine-tuning dataset, which includes six key features: solar irradiance, ambient temperature, PV output power, total output power, energy storage output, and the state of charge (SOC) of the battery. Solar irradiance and temperature directly influence the power generation of the PV system. The PV unit and the energy storage system operate in a coordinated manner where the PV unit generates electricity unidirectionally based on available sunlight, while the energy storage system enables bidirectional regulation. Excess energy charges the battery and increases the SOC when PV output exceeds load demand. Conversely, the energy storage system discharges to compensate for deficits when PV output falls short of demand, leading to a decrease in SOC. Fig. 7 illustrates the

graphical structure representing these abstracted causal relationships.

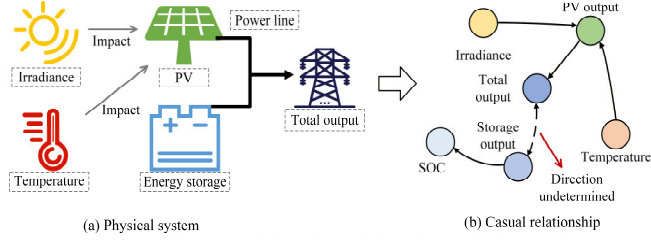


Fig. 7. (a) The influence relationship of the physical photovoltaic power plant. (b) The physics-statistics combined causal relationship derived from physical systems. The causal direction between the total output and storage output is identified through the causal discovery.

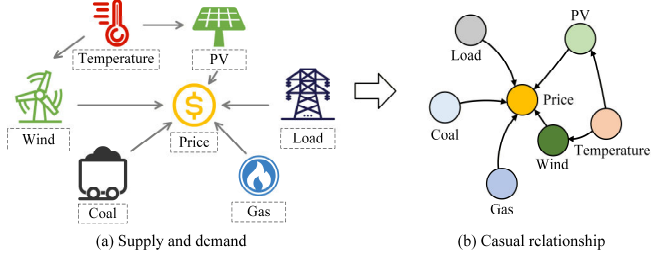


Fig. 8. (a) The influence relationship of the multi-energy system supply and demand. (b) The supply-demand combined causal relationship derived from market mechanisms.

Dataset B comprises hourly resolution data covering a broader spectrum of power system variables. The features include wind power, PV generation, electrical load, electricity prices, gas-fired generation, coal-fired generation, and ambient temperature. Prior domain knowledge defines specific causal relationships among these variables. Temperature acts as an external variable that influences both solar power generation and electrical load. The electricity price is modeled as a downstream variable influenced by the interplay of supply and demand. Specifically, renewable sources including solar and wind power, conventional generation from gas and coal units, and the system load all exert directed causal influence on the electricity price. These relationships are similarly abstracted into a directed graph structure to guide the model, as shown in Fig. 8.

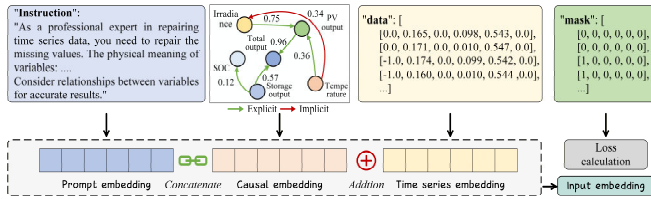


Fig. 9. A sample in the SFT dataset includes a prompt, a causal graph, a multivariate time series matrix and a mask matrix. The instruction shown in the figure is an abbreviated illustration; the complete task-specific prompt templates are provided in Appendix A. The causal graph provides a concrete construction example. The green edges represent explicit physical priors derived from domain laws in Fig. 7. The red edges represent implicit causal relationships discovered by statistical algorithms. The numerical values on the edges quantify the causal strengths calculated from the observational data.

A concrete sample under the imputation scenario is illustrated in Fig. 9. The prompt, serving as a prefix, incorporates role-playing and task instructions to specify the task, data types, and other contextual information, thereby enhancing the LLM's adaptability to downstream tasks. The causal relationships are represented as a directed weighted graph, where nodes denote variables and edges indicate the strength and direction of causal influences. The time series data and corresponding mask are arranged in two-dimensional matrices. After embedding, the time series form part of the model input. The model is optimized using a

composite objective function that combines both masked and global reconstruction losses. The mask matrix $\mathbf{M}$ is utilized to compute the masked loss, where element-wise multiplication restricts this specific loss component to the target regions. In parallel, an accuracy loss is calculated over the entire output. This combination ensures that the model achieves high accuracy on the specific task while preserving the overall temporal consistency of the generated time series.

TABLE II
TASK INFORMATION SUMMARY

Task	Dataset	Input resolution	Target / Horizon	Description
Imputation	A	1 min	Full sequence	Reconstruct missing values in PV/Storage data
	B	1 hour	Full sequence	Reconstruct missing values in multi-energy data
Forecasting	A	1 min	Hour-ahead	Ultra-short-term prediction
	A	1 min	Day-ahead	Short-term prediction
	B	1 hour	6-hour-ahead	Short-term prediction
Super-resolution	B	1 hour	Day-ahead	Short-term prediction
	A	15 min	5 min	3x upscaling of temporal resolution
	A	5 min	1 min	5x upscaling of temporal resolution
	B	2 hours	1 hour	2x upscaling from coarse measurements

Table II summarizes the scope of task types addressed in this paper. The evaluation covers three primary categories of time-series analysis tasks applied to the datasets. Imputation tasks address the reconstruction of missing values within a sequence. The model performs full sequence reconstruction but replaces only the missing entries with estimated counterparts. This approach preserves a fixed output length matching the original and avoids complications arising from variable-length sequences. Both dataset A and dataset B undergo testing for data imputation. Forecasting tasks vary by time horizon. For dataset A, the evaluation includes hour-ahead and day-ahead forecasting. For dataset B, the experiments extend to day-ahead prediction and a 6-hour short-term forecast. Super-resolution tasks enhance the granularity of coarse-grained data. Experiments on dataset A include upscaling 15-minute interval data to 5-minute resolution and 5-minute data to 1-minute resolution. Experiments on dataset B focus on enhancing 2-hour interval data to 1-hour resolution.

As comparisons, we train benchmarking models for every task: Multi-layer Perceptron (MLP), LSTM, Bidirectional LSTM (Bi-LSTM), Stacked Auto-Encoder (SAE), Transformer, DLinear [45], TimesNet [46], PatchTST [47], GPT4TS [12], TimeGPT [10].

All models are trained on NVIDIA A100 GPUs. The training process utilized the AdamW optimizer with a learning rate of 1×10^{-4} , a cosine annealing schedule to modulate the learning rate decay, a batch size of 16, and bfloat16 precision over 50 training epochs. To balance efficiency with performance, Low-Rank Adaptation (LoRA) is employed with hyperparameters tailored to the distinct characteristics of each dataset: for the high-resolution dataset A, a deeper fine-tuning strategy (rank=32, targeting all linear layers) is utilized to capture long-term dependencies ($L=2880$), whereas for the multi-energy dataset B, a lightweight configuration (rank=16) is adopted to accommodate shorter temporal dependencies ($L=48$). Regarding input modalities, the time-series data are embedded into vectors matching the hidden dimension of

the LLM backbone, while textual prompts are tokenized and mapped to the same high-dimensional space to ensure alignment between modalities.

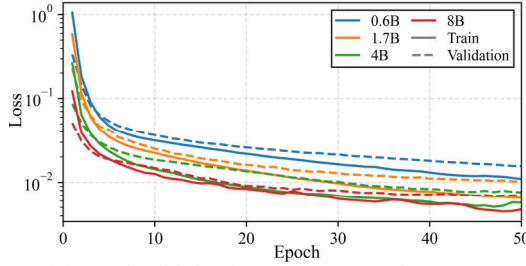


Fig. 10. Training and validation loss convergence of CM-LLM on Dataset A across different model scales (0.6B–8B) in logarithmic scale. Solid lines represent the training loss, while dashed lines denote the validation loss. The curves demonstrate rapid initial optimization and stable, asymptotic convergence without overfitting.

Fig. 10 illustrates the training and validation loss convergence curves for the CM-LLM variants (0.6B, 1.7B, 4B, and 8B) using a logarithmic scale on the y-axis. All model variants are trained with a fixed 50-epoch strategy to balance model accuracy and training cost. As shown, all model variants exhibit robust convergence behavior, with the losses decreasing rapidly within the first 10 epochs and then entering a stable, asymptotic decline in the later training stage. The gradual decrease after approximately 20 epochs is consistent with the cosine annealing learning rate schedule, under which smaller learning rates enable fine-grained parameter refinement near convergence. Notably, the validation loss closely tracks the training loss across all model scales without noticeable divergence, indicating that the models achieve stable convergence without evident overfitting and maintain reliable generalization performance.

B. Performance Comparison with Baselines

CM-LLM's performance was evaluated using both point-wise metrics to assess reconstruction accuracy and distribution-wise metrics to quantify statistical fidelity, including Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), Fréchet Inception Distance (FID) and Dynamic Time Warping (DTW).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (20)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (21)$$

$$FID = \|\mu_{\text{real}} - \mu_{\text{gen}}\|_2^2 + \text{Tr}(\Sigma_{\text{real}} + \Sigma_{\text{gen}} - 2(\Sigma_{\text{real}}\Sigma_{\text{gen}})^{1/2}) \quad (22)$$

$$DTW(R, G) = \min_w \sum_{s=1}^k d(w_s) \quad (23)$$

where y_i is the ground truth, $\hat{y}_i$ is the output data and n is the number of data points. In equation (22), μ is the mean of feature vectors, Σ is the covariance matrix and $\text{Tr}(\cdot)$ is the trace of a matrix. In equation (23), $d(w_s)$ is the distance between two points, and $w_s=(i, j)$ represents the correspondence between the i -th point of real data R and the j -th point of generated data G .

As summarized in Table III and IV, the CM-LLM demonstrates superior performance over established baselines across most tasks. The most significant gains are observed in imputation, attributed to the effective leveraging of causal analysis, and in day-ahead forecasting, confirming its reliable long-term predictive capability. While the model shows a negligible performance gap in simpler hour-ahead tasks and its super-resolution advantage diminishes at finer

TABLE III
RESULTS OF MODEL PERFORMANCE FOR DIFFERENT TASKS

Model	Metrics	Dataset A						Dataset B					
		Imputation		Forecasting		Super-resolution		Imputation		Forecasting		Super-resolution	
		all points	masked	day-ahead	hour-ahead	15m→5m	5m→1m	all points	masked	day-ahead	6-hour-ahead	2h→1h	
MLP	MAE	0.1033	0.1288	0.0996	0.0455	0.0356	0.0571	0.1211	0.1297	0.1363	0.1182	0.0838	
	RMSE	0.1583	0.1912	0.1893	0.0970	0.0508	0.0733	0.1752	0.1922	0.1884	0.1538	0.1162	
LSTM	MAE	0.0501	0.1231	0.1118	0.0874	0.0347	0.0576	0.0988	0.1232	0.1544	0.1404	0.0838	
	RMSE	0.1518	0.1847	0.2189	0.0951	0.0591	0.0838	0.1357	0.1747	0.2059	0.1800	0.1636	
Bi-LSTM	MAE	0.0416	0.0989	0.1261	0.0827	0.0283	0.0470	0.1044	0.1244	0.1528	0.1365	0.0962	
	RMSE	0.0805	0.1579	0.2259	0.1230	0.0535	0.0713	0.1239	0.1427	0.2053	0.1752	0.0962	
SAE	MAE	0.0893	0.1133	0.0910	0.0432	0.0361	0.0612	0.1200	0.1311	0.1388	0.1225	0.0801	
	RMSE	0.1367	0.1879	0.1708	0.0951	0.0525	0.0779	0.1802	0.2041	0.1910	0.1609	0.1120	
Transformer	MAE	0.0363	0.0859	0.1210	0.0832	0.0383	0.0624	0.0985	0.1033	0.1535	0.1401	0.0758	
	RMSE	0.0799	0.1453	0.1780	0.1251	0.0561	0.0818	0.1271	0.1631	0.2063	0.1785	0.1112	
DLinear	MAE	0.0854	0.0825	0.0791	0.0411	0.0355	0.0504	0.1332	0.1534	0.1344	0.1213	0.0690	
	RMSE	0.1269	0.1396	0.1518	0.0916	0.0552	0.0713	0.1749	0.2035	0.1667	0.1616	0.1077	
TimesNet	MAE	0.0530	0.1139	0.0944	0.0377	0.0340	0.0672	0.1251	0.1020	0.1632	0.1240	0.0758	
	RMSE	0.1295	0.1637	0.1660	0.0864	0.0578	0.0799	0.1752	0.1919	0.2058	0.1430	0.1079	
PatchTST	MAE	<u>0.0359</u>	<u>0.0383</u>	<u>0.0709</u>	0.0477	0.0294	<u>0.0463</u>	0.0920	0.1053	0.1109	<u>0.0777</u>	0.0513	
	RMSE	0.0793	<u>0.0801</u>	<u>0.1191</u>	0.0891	0.0461	0.0694	<u>0.1152</u>	0.1642	0.1509	<u>0.1120</u>	<u>0.0744</u>	
GPT4TS	MAE	0.0395	0.0391	0.1010	0.0451	<u>0.0281</u>	0.0488	0.0985	<u>0.0904</u>	<u>0.1082</u>	0.0980	0.0494	
	RMSE	<u>0.0764</u>	0.0829	0.1425	0.0893	0.0497	<u>0.0691</u>	0.1501	<u>0.1301</u>	<u>0.1459</u>	0.1316	0.0792	
TimeGPT	MAE	0.0827	0.1083	0.1060	0.0569	0.0401	0.0548	0.1108	0.1207	0.1374	0.1255	0.0747	
	RMSE	0.1389	0.1533	0.1529	0.1026	0.0590	0.0764	0.1669	0.1693	0.1703	0.1608	0.0938	
CM-LLM (0.6B)	MAE	0.0355	0.0368	0.0649	<u>0.0391</u>	0.0252	0.0458	<u>0.0948</u>	0.0893	0.1010	0.0751	<u>0.0497</u>	
	RMSE	0.0731	0.0748	0.1045	<u>0.0886</u>	<u>0.0470</u>	0.0685	0.1144	0.1242	0.1425	0.1093	0.0694	
Improvement	MAE	1.11%	3.92%	8.46%	-3.71%	10.32%	1.08%	-3.04%	1.22%	6.65%	3.35%	-0.61%	
	RMSE	4.32%	6.62%	12.26%	-2.55%	-1.95%	0.87%	0.69%	4.53%	2.33%	2.41%	6.72%	

*Bold term indicates the best performance, while underlining term represents second best.

TABLE IV
RESULTS OF MODEL PERFORMANCE FOR DIFFERENT TASKS

Metrics	FID						DTW					
Model	Irradiance	Temperature	PV output	Total output	Storage output	SOC	Irradiance	Temperature	PV output	Total output	Storage output	SOC
MLP	8.0628	1.5941	4.4366	8.8197	6.3652	4.5849	1.6512	0.8430	1.7778	1.4567	1.3679	0.6782
LSTM	5.4604	1.2371	3.8346	7.9281	6.2505	2.9319	1.6179	0.5584	1.3936	1.3152	1.3042	0.6861
Bi-LSTM	4.9163	0.6426	2.9828	4.7414	4.4824	2.2645	1.3034	0.5479	1.3476	1.1995	1.1902	0.6557
SAE	7.3341	1.5514	4.0297	7.1753	6.5175	3.9521	1.8810	0.6325	1.6184	1.5104	1.3700	0.8559
Transformer	5.3947	0.7587	2.9970	3.8391	4.0381	2.4570	1.5209	0.7966	1.3651	1.7234	1.6164	0.7394
DLinear	6.6924	1.5941	3.6503	7.4525	4.6791	2.9954	1.3958	0.5391	1.0792	1.3759	1.3801	0.5682
TimesNet	6.8100	1.3078	4.2114	6.2365	4.3066	2.6179	1.3513	1.3749	1.1775	1.5118	1.3205	0.8983
PatchTST	4.6262	<u>0.5459</u>	4.4366	<u>3.3906</u>	<u>3.7681</u>	<u>2.1983</u>	<u>0.8892</u>	<u>0.4545</u>	0.9895	<u>1.1927</u>	1.2414	0.5344
GPT4TS	<u>4.0017</u>	0.6113	<u>2.8859</u>	3.9728	3.8525	2.2591	0.9101	0.5979	<u>0.9445</u>	1.2875	1.6014	<u>0.5216</u>
TimeGPT	5.2123	1.4082	3.7410	5.9616	4.9614	3.7137	1.1644	0.6970	1.7070	1.5044	1.3856	0.7776
CM-LLM(0.6B)	3.7903	0.5228	2.8796	3.2489	3.7476	2.1845	0.8418	0.4022	0.9314	1.1447	<u>1.2217</u>	0.5128
Improvement	5.28%	4.23%	0.22%	4.18%	0.54%	0.64%	5.33%	11.51%	1.39%	4.02%	-2.65%	1.69%

*Bold term indicates the best performance, while underlining term represents second best.

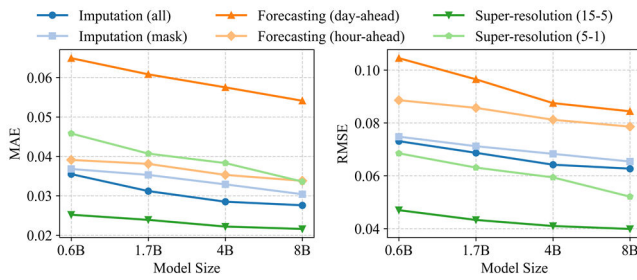


Fig. 11. Performance of different model sizes in tasks.

temporal resolutions, this is deemed an acceptable trade-off. The unified architecture inherently sacrifices some task-specific optimization to achieve broad applicability, demonstrating overall robustness in capturing complex temporal dependencies. In the super-resolution task, the method's superiority is validated by lower FID and DTW values, indicating strong fidelity in reconstructing data distributions and preserving temporal dynamics. Notably, the proposed model maintains its performance advantage for highly stochastic variables (e.g., solar irradiance). This robust performance in challenging scenarios further validates the model's advanced capability.

C. Performance Comparison with Different Model Sizes

Scaling laws suggest that larger language models tend to achieve better performance. To investigate this relationship within our framework, we conduct experiments using four model sizes from the Qwen3 series: 0.6B, 1.7B, 4B, and 8B parameters. The training results, shown in Fig. 11, provide insights into how model capacity influences performance under our setup. Both MAE and RMSE metrics exhibit a consistent downward trend as the model size increases. Specifically, for the MAE metric, the error decreases monotonically across all tasks as the model size grows from 0.6B to 8B parameters. For instance, the MAE for forecasting (day-ahead) drops from approximately 0.07 at 0.6B parameters to around 0.05 at 8B parameters. Similarly, the RMSE shows a parallel decrease, with the largest reduction observed in the forecasting (day-ahead) task, where the RMSE declines from about 0.10 at 0.6B parameters to roughly 0.08 at 8B parameters. This trend indicates that increasing the model capacity leads to improved performance across various tasks, which is consistent with the scaling laws hypothesis. The diminishing returns observed as the model size increases from 4B to 8B

parameters suggest that while larger models continue to offer performance gains, the marginal improvements become smaller. This finding underscores the importance of balancing model size with computational resources and highlights the potential for further optimization in model architecture and training strategies to maximize performance gains with increasing model capacity.

D. Ablation Study

To better understand the contribution of each model component, we conduct ablation studies by systematically removing specific elements and evaluating the resulting performance. In particular, we focus on the influence of prompt data and causal relationship graphs. For illustrative purposes and to maintain experimental clarity, all analyses are carried out using the Qwen3-0.6B.

To evaluate model fidelity beyond point-wise accuracy, we assess its ability to preserve two critical structural properties of the multivariate time series data: statistical inter-dependencies among variables and inherent physical constraints of the system. To this end, we introduce two specialized metrics:

1. **Correlation Discrepancy D** : This metric quantifies the model's capacity to replicate the statistical dependencies among variables. It is calculated as the sum of the absolute differences between the elements of the Pearson correlation matrix derived from the model output and that from the ground truth. A lower value indicates a more accurate representation of the inter-variable relationships.

2. **Power Balance Mean Absolute Error P_{MAE}** : This metric assesses the model's adherence to the physical law of power conservation (i.e., $Output = PV + Storage$). It is computed as the MAE between the power balance residual series derived from the model's predictions and that from the ground truth.

$$D = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left| \text{Corr}(x_i, x_j) - \text{Corr}(x_i, x_j) \right| \quad (24)$$

$$P_{MAE} = \frac{1}{n} \sum_{i=1}^n |R_i^{true} - R_i^{pred}| \quad (25)$$

$$R_i = P_i^{total} - (P_i^{PV} + P_i^{Storage}) \quad (26)$$

where (x_i, x_j) is the feature pair, $\mathbf{X}_o$ is the output data, $\mathbf{X}_g$ is the ground truth, Corr means the correlation coefficient calculation function, N is the number of feature and n is the number of data points. The results are summarized in Table

TABLE V
RESULTS OF ABLATION STUDY

Task	Model	None				Prompt-only				Causal-only				Proposed			
	Metric	MAE	RMSE	D	P_{MAE}	MAE	RMSE	D	P_{MAE}	MAE	RMSE	D	P_{MAE}	MAE	RMSE	D	P_{MAE}
Imputation	all points	0.043	0.088	2.340	0.058	<u>0.038</u>	<u>0.076</u>	2.321	0.050	0.039	0.078	2.236	<u>0.049</u>	0.036	0.073	<u>2.253</u>	0.047
	masked	0.049	0.083			0.040	0.077			<u>0.039</u>	0.073	<u>2.236</u>	<u>0.049</u>	0.037	0.075	<u>2.253</u>	0.047
Forecasting	day-ahead	0.087	0.142	1.638	0.038	<u>0.073</u>	<u>0.116</u>	1.527	<u>0.033</u>	0.074	0.132	<u>1.381</u>	0.036	0.065	0.104	1.138	0.026
	hour-ahead	0.053	0.097	2.523	0.029	0.037	0.083	2.346	0.027	0.050	0.092	<u>2.278</u>	0.015	<u>0.039</u>	<u>0.089</u>	2.114	<u>0.016</u>
Super-resolution	15min→5min	0.046	0.047	1.454	0.023	<u>0.031</u>	0.044	1.386	0.018	0.042	0.058	<u>1.139</u>	<u>0.016</u>	0.025	<u>0.047</u>	0.875	0.013
	5min→1min	0.051	0.079	2.317	0.031	<u>0.047</u>	<u>0.070</u>	2.194	0.029	0.048	0.072	<u>1.900</u>	0.022	0.046	0.068	1.815	<u>0.024</u>

*Prompt-only: the proposed method without causal graphs. Causal-only: the proposed method without prompt data. Bold term indicates the best performance, while underlining term represents second best.

V.

The results demonstrate that the causal graph and prompt data modules have distinct, complementary functions. The causal graph module is instrumental in capturing the system's structural and physical properties, systematically reducing D and P_{MAE} across all tasks; for instance, in day-ahead forecasting, it lowered the D value from 1.527 to 1.138. Concurrently, the prompt data module primarily improves predictive accuracy. For example, in the day-ahead forecasting task, adding prompt information to the causal-only variant reduces the MAE from 0.074 to 0.065 and the RMSE from 0.132 to 0.104. This synergy validates the integrated architecture: the causal graph enforces physical plausibility via structural regularization, while the prompt data infuses rich, dynamic information to maximize precision, yielding a model that is simultaneously highly accurate and physically reliable.

E. Impact of Fusion Mechanisms

To evaluate the efficacy of the proposed element-wise addition for integrating graph and temporal embeddings, a comparative analysis is conducted against concatenation, gating, and cross-attention mechanisms. As presented in Table VI, the element-wise addition strategy consistently yields the lowest RMSE and MAE across both datasets.

In terms of computational resources, addition, concatenation, and gating demonstrate comparable memory efficiency and inference throughput. In contrast, the cross-attention mechanism induces a substantial memory overhead of approximately 33% on dataset A without delivering improvements in predictive accuracy. Although the Gating mechanism achieves a marginal advantage in throughput on dataset A, it results in a notable increase in forecasting error compared to the addition strategy.

TABLE VI

COMPARISON OF DIFFERENT FUSION STRATEGIES

Dataset	Fusion Method	MAE ↓	RMSE ↓	Peak Memory (MB) ↓	Throughput (samples/s) ↑
Dataset A	Add	0.0874	0.0946	13,770	6.75
	Concat	0.0898	0.1021	<u>13,869</u>	6.70
	Gating	0.1055	0.1138	13,916	6.89
	Cross-Attn	0.1130	0.1337	18,336	6.47
Dataset B	Add	0.0860	0.0961	34,022	150.86
	Concat	0.1391	0.1028	34,050	147.55
	Gating	0.1141	0.0989	34,062	150.74
	Cross-Attn	<u>0.1012</u>	<u>0.0969</u>	46,066	137.60

*Bold term indicates the best performance, while underlining term represents second best.

The effectiveness of the additive strategy aligns with the geometric properties of high-dimensional latent spaces [48]. Due to the approximate orthogonality of high-dimensional vectors, element-wise addition functions as a superposition operator, enabling the preservation of distinct temporal and spatial information within shared subspaces without

destructive interference. Facilitated by learnable linear projections that ensure semantic alignment [49], this approach offers the optimal trade-off between predictive accuracy and computational efficiency.

F. Sensitivity Analysis of Prompt

To evaluate the efficacy of the proposed prompt construction strategy, we design four distinct configurations for sensitivity analysis on both dataset A and dataset B: a minimal baseline containing only core instructions; a paraphrased variant to test semantic robustness; a standard setting incorporating structured role definitions and physical context; and a dynamic configuration that further integrates real-time statistical features and trend descriptions.

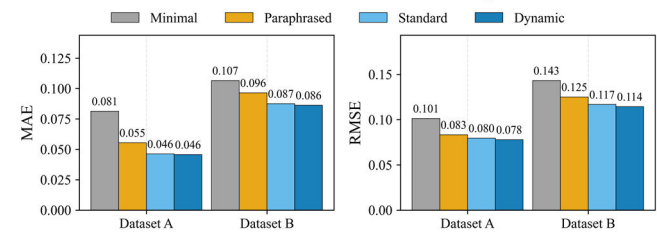


Fig. 12. Performance of different prompt construction strategies.

Fig. 12 presents the quantitative comparison of error metrics across these configurations. The minimal setting yields the highest error rates in all cases, underscoring the limitations of generic instructions lacking domain context. In contrast, the standard configuration achieves significant performance gains, with the mean absolute error on dataset A decreasing from 0.0813 to 0.0462. This substantial reduction indicates that incorporating role definitions and static physical context effectively activates the domain knowledge of the model. Furthermore, the dynamic configuration exhibits the lowest error metrics, demonstrating that integrating real-time statistical features provides explicit guidance that further enhances reconstruction accuracy. Notably, the paraphrased setting performs worse than the standard configuration. This performance degradation suggests that unstructured semantic variations may introduce unnecessary noise, whereas a fixed, structured prompt format is essential for maintaining consistency and aligning the model with the specific task requirements. Consequently, the consistent superiority of the standard and dynamic configurations confirms the necessity of a standardized, statistically aware prompt strategy for complex time-series analysis tasks.

G. Cost Analysis

There is a trade-off between model parameter scale and training cost. As summarized in Table VII, increases in parameter count consistently correlate with longer training durations, an effect attributed to greater operational complexity and memory consumption. A comparative

analysis of the Prompt-only and Causal-only approaches reveals distinct scalability profiles. While the Prompt-only (tokenizer-based) method incurs lower overhead at smaller scales, the GNN-based Causal-only variant, though initially more demanding, demonstrates superior training efficiency at large scales (e.g., 4B and 8B). This suggests that the computational overhead of the GNN module remains relatively stable, whereas the cost associated with the tokenizer embedding scales substantially with model size. Consequently, the Causal-only architecture exhibits better overall scalability for large-scale implementations.

TABLE VII
MODEL TRAINING COST

Model		Training time (h)	Training parameter	Training parameter percentage(%)	Latency (ms)
0.6B	None	1.86	21248006	3.4362	23.96
	Prompt-only	2.64			35.25
	Causal-only	3.28			150.10
	Proposed	4.58			158.83
1.7B	None	3.05	36983814	2.1017	39.88
	Prompt-only	4.16			55.26
	Causal-only	4.48			167.09
	Proposed	5.56			186.52
4B	None	6.17	68706822	1.6783	84.18
	Prompt-only	8.89			119.57
	Causal-only	8.06			238.57
	Proposed	9.72			271.99
8B	None	9.08	91524102	1.1045	133.90
	Prompt-only	11.81			181.60
	Causal-only	10.83			290.23
	Proposed	12.67			338.26

In addition to training costs, inference latency is evaluated to assess the feasibility for real-time operations, as detailed in the last column of Table VII. During inference, an individual causal graph is dynamically generated before the data enters the language model. Empirical profiling demonstrates that this graph generation requires an average of 6.25 milliseconds for Dataset A and 2.28 milliseconds for Dataset B on a standard processor. This overhead is extremely small. The total inference time exhibits a positive correlation with model scale. The latency increases from 158.83 milliseconds for the 0.6B model to 338.26 milliseconds for the 8B version. The absolute duration remains below one second. Typical power system applications operate at a resolution of minutes. Therefore, a processing time of a fraction of a second is entirely acceptable. These results confirm that the proposed framework satisfies the operational time constraints of modern power grids.

V. CONCLUSION

In this paper, we propose a causal-guided multimodal large language model (CM-LLM) to achieve generalized power system time series analytics. A physics-statistics combined causal discovery mechanism is proposed to capture the complex causal relationship among power system time series variables, and a multimodal data preprocessing framework is designed to achieve data fusion to enhance the model accuracy. Besides, a generic “mask-and-reconstruct” paradigm is proposed to solve the task specificity issue, while a dynamic input-output padding mechanism is designed to handle the structural rigidity problem. CM-LLM can fully unleash the generalization capabilities of LLMs in solving heterogeneous power system time series analysis tasks with superior accuracy and efficiency.

Three different time series tasks, i.e., missing data imputation, forecasting and super resolution are selected as examples to evaluate the model performances. Experimental results demonstrate that even when using Qwen3-0.6B as the backbone model, CM-LLM significantly outperforms conventional baseline approaches. Moreover, as the scale of the pre-trained LLM increases, the performance improves accordingly. We further conduct ablation studies to assess the contributions of task prompts and causal graph information, both of which are shown to play a critical role in enhancing model performances. However, it should be noted that performance gains tend to diminish as the model size increases to a certain level, while the training costs continue to rise significantly. Such observations highlight the need for a balanced trade-off between model scalability and efficiency. Regarding fusion mechanisms, comparative experiments validate the superiority of the element-wise addition strategy, which achieves the optimal trade-off between accuracy and efficiency. Furthermore, sensitivity analysis confirms that structured prompts enriched with statistical features significantly enhance performance, whereas unstructured semantic variations tend to introduce noise.

Future work will focus on extending the framework's applicability to a broader range of task types and data formats, thereby achieving a more generalized processing paradigm.

APPENDIX

A. Prompt Format Used in CM-LLM

To improve reproducibility, the prompt templates used in this work are provided below in textual form. $\{Task\ Type\}$, $\{Statistical\ Description\}$, and $\{Time\ Series\ Data\ Sequence\}$ are placeholders instantiated according to the specific task and input sample.

B. Prompt Template for Dataset A

[Role Prompting] You are an expert in renewable energy systems, battery energy storage operations, and time-series data analysis.

[Task Type] Your current task is to perform time-series $\{Task\ Type\}$ (e.g., Imputation / Forecasting / Super-resolution).

[Data Scenario] The data is collected from a real-world photovoltaic (PV) power plant equipped with an energy storage system at a 1-minute temporal resolution. The input sequences contain 6 variables in the following ordered sequence: [Solar Irradiance, Ambient Temperature, PV Output Power, Total Output Power, Storage Output, State of Charge]. Please strictly adhere to the following physical and causal relationships: Generation: Solar Irradiance and Ambient Temperature directly govern the PV Output Power. Aggregation: The Total Output Power is jointly determined by the unidirectional PV Output Power and the bidirectional Energy Storage Output. Regulation: The Energy Storage system operates bidirectionally. Charging (storing excess PV energy) increases the SOC, while discharging (compensating for deficits) decreases the SOC.

[Statistical Input Analysis] Based on the historical observation window, the statistical characteristics of the current numerical features are as follows: $\{Statistical\ Description\}$.

[Data Input & Instruction] The raw time-series sequence is provided below. Note that the target values you need to predict or impute are explicitly masked and marked as -1 in the sequence. $\{Time\ Series\ Data\ Sequence\}$ Please execute the $\{Task\ Type\}$ task by analyzing the context and output the predicted numerical values for the -1 positions directly.

C. Prompt Template for Dataset B

[Role Prompting] You are an expert in power economics, regional grid dispatch, and energy market time-series analysis.

[Task Type] Your current task is to perform time-series *{Task Type}* (e.g., Imputation / Forecasting / Super-resolution).

[Data Scenario] The data represents a regional energy market driven by supply and demand dynamics. The input sequences contain 7 variables in the following ordered sequence: [Temperature, Wind Power Generation, PV Power Generation, Coal Power Generation, Gas Power Generation, Load Demand, Electricity Price]. Please strictly adhere to the following physical and economic causal relationships: Meteorological Impact: Ambient Temperature directly influences the generation capacity of renewable energy sources (Wind and PV). Market Pricing Mechanism: The Electricity Price serves as the central hub variable. It is dynamically determined by the balance between the aggregate power supply (Coal, Gas, Wind, PV) and the power demand (Load). Supply-Demand Interactions: Fluctuations in uncontrollable renewable generation (Wind/PV) or Load demand trigger adjustments in dispatchable generation (Coal/Gas), subsequently driving Electricity Price volatility.

[Statistical Input Analysis] Based on the historical observation window, the statistical characteristics of the current numerical features are as follows: *{Statistical Description}*.

[Data Input & Instruction] The raw time-series sequence is provided below. Note that the target values you need to predict or impute are explicitly masked and marked as -1 in the sequence. *{Time Series Data Sequence}* Please execute the *{Task Type}* task by analyzing the context and output the predicted numerical values for the -1 positions directly.

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